

Data Analysis and visualization using excel

Assignment -2

Last date of submission 10 -Feb-2026

(Hard copy handwritten)

1. Define data visualization. Explain its importance in the modern data-driven ecosystem, particularly in supporting decision-making processes across organizations.
2. Discuss the fundamental principles of effective data visualization. How do these principles help in reducing cognitive load and improving data interpretation?
3. Differentiate between exploratory and explanatory data visualization. Provide suitable examples to illustrate when each type is used.
4. Explain the role of charts and graphs in data communication. Why is graphical representation often preferred over tabular data?
5. Describe at least five commonly used types of charts and graphs. For each, explain the kind of data it is best suited to represent.
6. What factors should be considered when choosing the right visualization for a given dataset? Discuss the consequences of selecting an inappropriate visualization.
7. Explain the complete data lifecycle covering data generation, collection, storage, processing, and analysis. Highlight the interdependence of these stages.
8. Discuss various sources of data generation. How does the nature of data (structured, semi-structured, unstructured) influence collection and storage strategies?
9. Examine the importance of data storage and processing in large-scale data systems. How do these stages impact the efficiency of data analysis?
- 10 Define the role of a Data Engineer. Discuss the key responsibilities of a Data Engineer in enabling effective data visualization and analytics.
- 11 Describe the ETL (Extract, Transform, Load) process. Explain its importance in data processing and transformation pipelines.
- 12 Compare OLAP and OLTP systems. Highlight their applications in analytical and transactional environments.

- 13 Explain Data Modeling concepts. What is an E-R Diagram? Describe its components with examples.
- 14 Explain basic charts and advanced charts available in Excel. Differentiate between Combo Charts and Pivot Charts with examples
- 15 What is a dashboard? Explain the use of PivotTables and Slicers in creating interactive dashboards.
- 16 Explain the role of spreadsheets (Excel / Google Sheets) in data visualization. How are they useful for small and medium-scale data analysis?
- 17 Describe the complete data lifecycle including data generation, collection, storage, processing, and analysis with suitable examples
- 18 Explain different types of charts and graphs such as Column, Line, Pie, and Bar charts. Provide one real-world use case for each type.
- 19 What factors should be considered while choosing the right visualization technique? Explain how incorrect chart selection can mislead data interpretation
- 20 Explain different data storage systems such as Relational Databases, NoSQL Databases, and Data Warehouses. When should each be used?